Welcome To Colab - Colab

colab.research.google.com/notebooks/intro.ipynb#scrollTo=3kpeidILTaxv

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```python
print("Recall   :", recall_score(y_test, pred, average='weighted'))

# ------------------------------
# Feature Set 2 (All Features)
# ------------------------------
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42)

model = GaussianNB()
model.fit(X_train, y_train)

pred = model.predict(X_test)

print("\nFeature Set 2")
print("Accuracy :", accuracy_score(y_test, pred))
print("Precision:", precision_score(y_test, pred, average='weighted'))
print("Recall   :", recall_score(y_test, pred, average='weighted'))
```

```
Feature Set 1
Accuracy : 0.8222222222222222
Precision: 0.833841684822077
Recall   : 0.8222222222222222

Feature Set 2
Accuracy : 0.9777777777777777
Precision: 0.9793650793650793
Recall   : 0.9777777777777777
```

Variables   Terminal   ✓ 12:38 PM   Python 3

```python
mle_mean = np.mean(data)
mle_variance = np.var(data)

print("MLE Mean     :", mle_mean)
print("MLE Variance :", mle_variance)

# ------------------
# Bayesian Estimation
# Prior Mean = 8
# Prior Variance = 4
# ------------------
prior_mean = 8
prior_variance = 4

sample_variance = np.var(data)

posterior_mean = (
    (prior_mean/prior_variance + len(data)*mle_mean/sample_variance) /
    (1/prior_variance + len(data)/sample_variance)
)

print("Bayesian Mean:", posterior_mean)
```

```
MLE Mean     : 9.5
MLE Variance : 8.25
Bayesian Mean: 9.243523316062177
```